Xiao Lin

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Research Interest

I am a Computer Science Ph.D. student at the University of Illinois Urbana-Champaign, advised by Prof. Hanghang Tong. I am deeply engaged in advancing **controllable and reliable AI systems under dynamic environments**. Recently, I investigate **how to harness agentic reasoning** through the integration of **memory, agentic reinforcement learning, and multi-agent systems**.

Education

University of Illinois Urbana-Champaign — Champaign, IL, USA
CS Ph.D. Advisor: Dr. Hanghang Tong — *Aug. 2023 – 2027 (expected)*

Peking University — Beijing, China
CS B.E. Advisor: Dr. Yisen Wang — *Sep. 2019 – Jun. 2023*

Industry Experience

Research Intern — Menlo Park, CA, USA
Meta Recommendation System, Meta. Manager: Dr. Shuguang Ye. — *May. 2026 – Aug. 2026*
- Researched on long-horizon agentic reinforcement learning.

Research Intern — Sunnyvale, CA, USA
Core Machine Learning Team, Meta. Manager: Dr. Yajuan Wang. — *May. 2025 – Oct. 2025*
- Researched on Mixture of Experts for long-sequence recommendation.

Research Intern — Beijing, China
Mobile Intelligence Business Team, SenseTime. Manager: Dr. Tong Li. — *May. 2021 – Nov. 2021*
- Researched on human posture detection and facial image generation.

Publications

2026 *LLM Agents, Safety & Alignment*

P1. **Xiao Lin**, Philip Li, Zhichen Zeng, et al. ALERT: Zero-shot LLM Jailbreak Detection via Internal Discrepancy Amplification. EMNLP 2026.

P2. **Xiao Lin**, Zhining Liu, Ze Yang, et al. MORALISE: A Structured Benchmark for Moral Alignment in Visual Language Models. ICML 2026.

P3. Xuying Ning, Katherine Tieu, Dongqi Fu, . . . , **Xiao Lin**, . . . , Hanghang Tong, Jingrui He. Code as Agent Harness. Preprint.

P4. Tianxin Wei, Ting-Wei Li, Zhining Liu, . . . , **Xiao Lin**, . . . , Hanghang Tong, Jingrui He. Agentic Reasoning for Large Language Models: A Survey. Preprint.

P5. Tianxin Wei, Zhan Shi, Minhua Lin, . . . , **Xiao Lin**, . . . , Jingrui He, Hanqing Lu. Evo-Harness: Context-to-Harness Skill Compilation for Self-Evolving Agents. EMNLP 2026.

P6. Yuanchen Bei†, **Xiao Lin**†, Zihao Li, Zhichen Zeng, et al. Physics-Informed Graph Transformer for Power-Grid Systems. Under review.

P7. Yuanchen Bei, Tianxin Wei, Xuying Ning, Yanjun Zhao, Zhining Liu, **Xiao Lin**, Yada Zhu, et al. Mem-Gallery: Benchmarking Multimodal Long-Term Conversational Memory for MLLM Agents. ACL 2026.

P8. Shuhang Lin, Zhencan Peng, Lingyao Li, **Xiao Lin**, Xi Zhu, Yongfeng Zhang. Cache Mechanism for Agent RAG Systems. Preprint.

P9. Shuhang Lin, Chuhao Zhou, **Xiao Lin**, Zihan Dong, Kuan Lu, Zhencan Peng, Jie Yin, Dimitris Metaxas. Conformal Path Reasoning: Trustworthy Knowledge Graph Question Answering via Path-Level Calibration. ICML 2026.

P10. Zhichen Zeng, Qi Yu, **Xiao Lin**, Ruizhong Qiu, Xuying Ning, Tianxin Wei, Yuchen Yan, Jingrui He, Hanghang Tong. Harnessing Consistency for Robust Test-Time LLM Ensemble. EACL 2026.

P11. Zhining Liu, Tianyi Wang, **Xiao Lin**, Penghao Ouyang, Gaotang Li, Ze Yang, et al. Do VLMs Have a Moral Backbone? A Study on the Fragile Morality of Vision-Language Models. ACL Findings 2026.

P12. Tianxin Wei, Zhichen Zeng, ..., **Xiao Lin**, ..., Hanghang Tong, Jingrui He. Inference Scaling of LLM Ensembling: Bridging Token Spaces with Token Translation. COLM 2026.

P13. Ting-Wei Li, Yuanchen Bei, **Xiao Lin**, Hanghang Tong. Beyond LLM-Based Reasoning: Lightweight GNNs for Agent Failure Attribution. Under review.

P14. Xuying Ning, Dongqi Fu, Tianxin Wei, ..., **Xiao Lin**, Lecheng Zheng, Jingrui He. OTU: Process-aware Uncertainty Estimation for Multimodal Reasoning. Under review.

2025 *Multimodal Alignment, Sequence Modeling*

P15. **Xiao Lin**, Robert Tang, Weilin Cong, et al. Mixture of Sequence: Theme-Aware Mixture-of-Experts for Long-Sequence Recommendation. WWW 2026.

P16. **Xiao Lin**, Zhichen Zeng, Tianxin Wei, Zhining Liu, Hanghang Tong. CATS: Mitigating Correlation Shift for Multivariate Time Series Classification. Under review.

P17. Duo Zhou, Yuji Zhang, Tianxin Wei, Ruizhong Qiu, Ke Yang, **Xiao Lin**, Cheng Qian, Jingrui He, Hanghang Tong, Heng Ji, Huan Zhang. Geometric-Disentanglement Unlearning. EMNLP 2026.

P18. Zhichen Zeng, Wenxuan Bao, **Xiao Lin**, Ruizhong Qiu, et al. Subspace Alignment for Vision-Language Model Test-time Adaptation. EMNLP 2026.

P19. Zihao Li, **Xiao Lin**, Zhining Liu, Jiaru Zou, et al. Language in the Flow of Time: Time-Series-Paired Texts Weaved into a Unified Temporal Narrative. ICLR 2026.

P20. Zhichen Zeng, Mengyue Hang, Xiaolong Liu, Xiaoyi Liu, **Xiao Lin**, Ruizhong Qiu, Tianxin Wei, Zhining Liu, et al. Hierarchical LoRA MoE for Efficient CTR Model Scaling. Under review.

P21. Zhichen Zeng, Ruizhong Qiu, Wenxuan Bao, Tianxin Wei, **Xiao Lin**, Yuchen Yan, et al. Pave Your Own Path: Graph Gradual Domain Adaptation on Fused Gromov-Wasserstein Geodesics. Under review.

P22. Baoyu Jing, Sanhorn Chen, Lecheng Zheng, ..., **Xiao Lin**, Yuchen Yan, Qi Yu, Dongqi Fu, Jingrui He, Hanghang Tong. TSAQA: Time Series Analysis Question And Answering Benchmark. ACL 2026.

P23. Zihao Li[†], Zhichen Zeng[†], **Xiao Lin**[†], Feihao Fang, Yanru Qu, Zhe Xu, et al. Flow Matching Meets Biology and Life Science: A Survey. Nature Partner Journals Artificial Intelligence 2026.

P24. Zhining Liu, Ze Yang, **Xiao Lin**, Ruizhong Qiu, Tianxin Wei, et al. Breaking Silos: Adaptive Model Fusion Unlocks Better Time Series Forecasting. ICML 2025.

P25. Ruizhong Qiu, Jun-Gi Jang, **Xiao Lin**, Lihui Liu, Hanghang Tong. TUCKET: A Tensor Time Series Data Structure for Efficient and Accurate Factor Analysis over Time Ranges. VLDB 2025.

P26. Dongqi Fu, Yada Zhu, Zhining Liu, Lecheng Zheng, **Xiao Lin**, Zihao Li, Liri Fang, Katherine Tieu, et al. ClimateBench-M: A Multi-Modal Climate Data Benchmark with a Simple Generative Method. CIKM 2025.

2024 *Trustworthy AI, Structure Mining*

P27. **Xiao Lin**, Zhining Liu, Dongqi Fu, Ruizhong Qiu, Hanghang Tong. BackTime: Backdoor Attacks on Multivariate Time Series Forecasting. **NeurIPS 2024 Spotlight**.

P28. **Xiao Lin**, Jian Kang, Weilin Cong, Hanghang Tong. BeMap: Balanced Message Passing for Fair Graph Neural Network. Learning on Graphs Conference 2024.

P29. **Xiao Lin**, Mingjie Li, Yisen Wang. MADE: Graph Backdoor Defense with Masked Unlearning. ArXiv 2024.

Project Experiences

Methodology Advancements for Generalizable Insights into Complex Systems DARPA
Research Assistant *Spring 2026 – Present*

NetSafe: Towards a Computational Foundation of Safe Graph Neural Networks NSF
Research Assistant *Fall 2024 – Present*

Sequential User Understanding Meta
Research Assistant *Fall 2025*

Ethical Understanding of Vision Language Models Fidelity
Research Assistant *Spring 2025 – Present*

AI for Future Agricultural Resilience, Management, and Sustainability NIFA
Research Assistant *Fall 2023 – Fall 2024*

Safety Alignment of Large Language Models VISA Research
Research Assistant *Fall 2024 – Fall 2025*

Programming Fabric for Geospatial Representation, Fusion and Federation IBM
Research Assistant *Fall 2023 – Present*

Modeling Adversarial Activities DARPA
Research Assistant *Fall 2023 – Spring 2024*

Teaching Experience

Teaching Assistant Champaign, IL, USA
CS 412 Data Mining *Spring 2026*

Teaching Assistant Champaign, IL, USA
CS 410 Text Information Retrieval *Spring 2025*

Awards

A1. Research Gold Award 2026

A2. ICML Silver Reviewer Award 2026

A3. Outstanding Graduate Award 2023

A4. Academic Excellence Award 2022

A5. Third-class Scholarship 2022

A6. Academic Excellence Award 2021

A7. Excellent Scientific Research Award 2021

A8. Third-class Scholarship 2020

A9. First Prize in the Chinese Physics Olympiad 2020

A10. First Prize in the Chinese Physics Olympiad 2018

Service

Conference reviewer: ICML, ICLR, NeurIPS, CVPR, ECCV, KDD, CIKM, TheWebConf, DSAA, LOG.
Journal reviewer: ACM Computing Survey, TMLR, IEEE Transaction on Computational Biology and Bioinformatics.